3 June 1975
SUPERSEDING
MIL-C-4279C
27 August 1969

MILITARY SPECIFICATION

CLOTH, MUSLIN, COTTON (PARACHUTE CANOPY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of cotton muslin cloth.

1.2 Classification. The cloth shall be of the following types as specified (see 6.2):

Type II	-	3.5 ounces
Type III	-	3.6 ounces (Mildew resistant treated)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-276	-	Thread, Cotton
PPP-P-1134	-	Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

FSC 8305

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:28Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580).

3. REQUIREMENTS

3.1 Standard sample. The dyed and finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

* 3.2 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.3 and 6.5).

3.3 Materials. The warp and filling shall be singles yarns spun from carded and drawn cotton.

3.4 Color. The color of the dyed and finished cloth shall be as specified (see 6.2) and shall match the standard sample (see 6.3). When given shades are specified the shade shall be obtained by the use of direct dyestuffs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:28Z
Check the source to verify that this is the current version before use.

* 3.4.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.6. When a standard sample is not available, the dyed and finished cloth shall show not more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.6.

3.4.2 Matching. The color of the dyed and finished cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin, and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

* 3.4.3 Colorfastness. The dyed and finished cloth shall show no more crocking than the standard sample when tested as specified in 4.6. When no standard sample is available, the dyed and finished cloth shall show no more crocking than Munsell Value of 6.5 when tested as specified in 4.6.

* 3.5 Physical requirements. The dyed and finished cloth shall conform to the requirements in table I when tested as specified in 4.6.

TABLE I. Physical requirements

Type	Weight oz/sq. yd. Minimum	yarns/inch Minimum 1/		Breaking strength pounds Minimum		Tearing strength pounds Minimum		Air permeability cu./ft/min. sq. ft.	
		Warp	Filling	Warp	Filling	Warp	Filling	Min.	Max.
II	3.5	54	56	48	42	3.0	2.5	170	230
III	3.6	56	58	48	45	3.0	2.5	130	190

1/ A minus tolerance of four yarns per inch in either the warp or filling will be permitted provided there is a total of not less than 110 yarns per inch for type II, and 114 yarns per inch for type III.

3.5.1 Width. The width shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

3.6 Weave. The weave shall be plain.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:28Z
Check the source to verify that this is the current version before use.

3.7 Processing. The cloth shall be dyed without prior preparation except optional light scouring. Bleaching is not permitted. Dyeing shall be by padding with direct dyes at or near the boil. Finishing materials for type III cloth as described in 3.8 shall be incorporated in the padding solution. Tension in processing greater than necessary for control purposes shall be avoided.

3.8 Finish. The type III cloth shall be treated with a dispersed polyethylene softener, and for mildew resistance by application of salicylanilide in an amount not less than 0.5 percent and not more than 1.0 percent of the weight of the cloth. Application of softener and mildew inhibitor shall be in combination with the dyeing (see 3.7 and 6.4).

3.9 pH. The pH value of the water extract of the dyed and finished cloth shall be no less than 5.5 nor more than 8.5 when tested as specified in 4.6.

3.10 Seam efficiency. The finished cloth shall have a seam efficiency of not less than 90 percent when tested as specified in 4.6.

* 3.11 Length and put-up. Unless otherwise specified (see 6.2), the finished cloth shall be furnished in rolls, each not less than 40 yards. The cloth shall be put-up in rolls as specified in PPP-P-1134.

3.12 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.13 Marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the roll.

3.14 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards, when calculated as specified in section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:28Z
Check the source to verify that this is the current version before use.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. The first article submitted in accordance with 3.2 shall be visually inspected for appearance and color. The sample shall be tested for physical properties in accordance with methods specified in 4.6.

* 4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of 4.4.2.1 through 4.4.2.4.

* 4.4.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined and defects clearly noticeable at normal inspection distance (3 feet) shall be classified as listed. The defects found shall be counted regardless of their proximity to each other, except when two or more defects represent a single local condition of the cloth, only one shall be scored. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The sample unit shall be 1 yard, full width of the finished cloth. The acceptable quality level shall be 6.5 defects per 100 units (yards). The sample size shall be in accordance with inspection level I of MIL-STD-105. The lot size shall be expressed in units of one yard each. The number of rolls selected shall be in accordance with table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17 03:28Z
Check the source to verify that this is the current version before use.

Defects

Hole, cut or tear
Smash, multiple floats, grease or oil stain
Embedded crease or wrinkle in excess of 1/2 inch width
Broken or missing pick or end

* 4.4.2.1.1 Overall examination. Each defect shall be counted no more than once in each roll examined. The sample unit shall be one roll. The size (number of rolls selected as sample) and the acceptance number shall be as shown in table II.

Defects

Width less than specified 1/
Overall uncleanness 1/
Baggy, ridgy or wavy cloth

1/ Shall not be scored unless it occurs for more than 5 yards per roll.

TABLE II. Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number

1/ If a lot contains fewer than three rolls, each roll in the lot shall be

4.4.2.2 Examination for length.

4.4.2.2.1 Individual rolls. During the yard-by-yard examination each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:28Z
Check the source to verify that this is the current version before use.

4.4.2.3 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Examination for face markings and compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have the face stamping missing from either or both ends or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.5 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking comply with section 5 requirements.

* 4.6 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be 4 continuous yards (full width) of the finished cloth. The sample size (number of sample units) shall be as shown below. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The sample size shall be as follows:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. Test methods

Characteristic	Requirement paragraph	Test method
Material (cotton yarn)	3.3	1200 1/
Presence of labile sulfur	3.4.1	2020
Colorfastness to crocking	3.4.3	5651
Weight	3.5	5041
Dye identification	3.4	1/

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:28Z
Check the source to verify that this is the current version before use.

TABLE III. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Yarns per inch		
Warp	3.5	5050
Filling	3.5	5050
Breaking strength		
Warp	3.5	5100
Filling	3.5	5100
Tearing strength		
Warp	3.5	5132
Filling	3.5	5132
Army permeability	3.5	5450
Weave	3.6	Visual 2/
Processing	3.7	1/
Polyethylene softener (type III)	3.8	1/
Salicylanilide content (type III)	3.8	1/
pH	3.9	2811
Seam efficiency	3.10	5110 3/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination per sample unit and the result reported as "pass" or "fail".

3/ The needle shall measure 0.040 inch (plus or minus) 0.001 inch across the blade at the eye. The cotton thread shall be type IA1, ticket No. 40, 3 ply in the needle, and ticket No. 60, 3 ply in the looper, conforming to V-T-276.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A or C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:28Z
Check the source to verify that this is the current version before use.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in cargo parachutes.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type required (see 1.2).
c. Width of cloth required (see 3.5.1).
d. Color of cloth required (see 3.4).
e. First article. When preproduction sample is not required (see 3.2).
f. Minimum length if other than specified (see 3.11).
g. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids or request for proposals.

6.4 A practical dyeing procedure for the type III cloth is as follows: The typical dyebath consists of direct dyestuffs, 2 percent of a polyethylene-sewing aid compound; 5-10 percent common salt; 1.5 percent of salicylanilide if mildew resistance is required. Material is padded at 160° - 180°F. (preferably at the latter temperature) to achieve maximum color value and required colorfastness properties. Drying of the cloth is conducted at such speeds as to minimize migration but at the same time to permit the further fixation of the dye in the early stages as the dye is concentrating within the cloth. The foregoing is for information only and is not mandatory.

* 6.5 First article. When a first article is required it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of 4 yards full width of the finished cloth. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:28Z
Check the source to verify that this is the current version before use.

6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - WC
Navy - MC
Air Force - 45, 82

User activity:

Army - AV

Preparing activity:

Army - GL

Project No. 8305-0127

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-4279D - CLOTH, MUSLIN, COTTON (PARACHUTE CANOPY)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:28Z
Check the source to verify that this is the current version before use.